

Technology Stack

Date	26 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML, CSS	Provides a simple and user-friendly web interface for applicants to enter loan details and view loan approval prediction results. CSS is used for designing and improving the visual appearance of the application pages.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Python, Flask	Flask handles HTTP requests, receives applicant inputs, connects the trained Machine Learning model, performs predictions, and returns loan approval or rejection results with confidence score.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	CSV Dataset, Joblib (.pkl files)	The historical loan dataset is stored in CSV format (loan_prediction.csv). The trained ML model (best_loan_model.pkl) and preprocessing objects (preprocessed_loan_data.pkl) are stored using Joblib for prediction during application execution. No traditional database is used in the current prototype.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render Cloud Platform	The application is deployed on Render and accessible through a hosted URL. The Flask-based Smart Lender application runs on the cloud platform, allowing users to access the loan prediction system online.
5	Version Control & CI/CD <i>(e.g., GitHub, GitLab, Jenkins)</i>	Git, GitHub	Git is used for version control and tracking project changes. GitHub is used for storing source code, collaboration among team members, and maintaining project history.

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Imbalanced-learn	These libraries are used for data analysis, visualization, preprocessing, class balancing, model training, evaluation, and loan approval prediction. XGBoost is used as the final best-performing classification model.
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